Parallel projection in space

How a three-dimensional figure is drawn on a flat page — what survives the projection and what does not.

LESSON 27–28 | 2 × 40 MIN | GEOMETRY 10, §13 | IGCSE E14.X

LESSON CLOCK

10' | 22' | 22' | 24' | 16' | 6'

The construction	10 min
What is preserved	22 min
What is not	22 min
Drawing solids correctly	24 min
Practice	16 min
Homework	6 min

LEARNING OBJECTIVES

- Construct the parallel projection of a point, segment and figure onto a plane.
- State the properties preserved by parallel projection.
- Explain why a circle projects to an ellipse and a square to a parallelogram.
- Draw a correct picture of a cube, prism and pyramid.

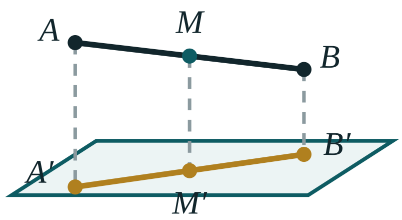
TEXTBOOK

National	pp. 125–136
Cambridge	Core & Extended, pp. 210–216
Workbook	Exercise 13.1

01 Explanation

The construction

Fix a plane α and a direction ℓ not parallel to it. Through each point A of the figure draw the line through A parallel to ℓ; where it meets α is the **projection** A'.



the midpoint projects to the midpoint

Every point drops along the same direction. The midpoint lands on the midpoint.

WHY IT MATTERS

Every picture of a solid on paper — every diagram in this book — is a parallel projection. Knowing what it preserves is knowing what you may legitimately read off a drawing.

What is preserved

PROPERTY	PRESERVED?
a straight line projects to a straight line (or a point)	yes
parallel lines project to parallel lines	yes
the ratio in which a point divides a segment	yes
the midpoint of a segment	yes
the ratio of lengths on **parallel** lines	yes
lengths	**no**
angles	**no**
the ratio of lengths on non-parallel lines	**no**

NEVER MEASURE A DRAWING

A right angle in the solid may appear as $60°$ on the page, and two equal edges may appear to differ. A drawing is a guide to structure, never to size.

What the images look like

FIGURE	ITS PARALLEL PROJECTION
square, rectangle, rhombus	a parallelogram (any one)
parallelogram	a parallelogram
triangle	a triangle (any one)
circle	an ellipse
trapezium	a trapezium — the parallel pair stays parallel

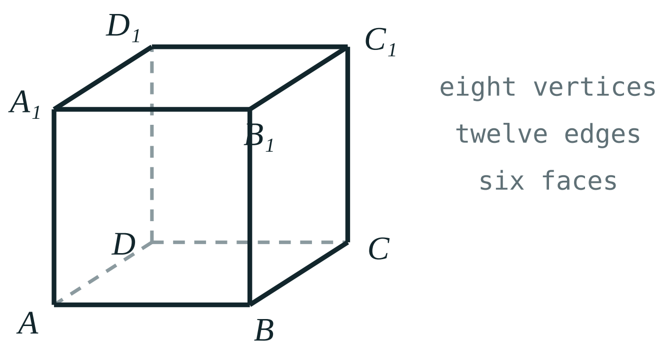
Because **any** triangle can be the image of **any** triangle, and any parallelogram the image of any square, a picture of a cube may be drawn with the base as any convenient parallelogram. That freedom is what makes cabinet projection — the drawing convention of this book — legitimate.

Drawing solids correctly

THE RULES OF A CORRECT PICTURE

1 Parallel edges are drawn parallel. **2** Midpoints are drawn at midpoints. **3** Hidden edges are dashed. **4** The base is drawn as a parallelogram, not as a rectangle. **5** Vertical edges stay vertical.

The standard convention here is **cabinet projection**: the depth axis is drawn at $45°$ and at half scale, so a unit cube appears with a receding edge of length 0.5. It is not what the eye sees, but it makes every parallel edge parallel on the page, which is what a geometry diagram needs.



A correct cube: parallel edges parallel, hidden edges dashed, base a parallelogram.

02 Worked examples

EXAMPLE 1 **M is the midpoint of AB. Where is M' on $A'B'$?**

1. Parallel projection preserves the ratio of division.
2. $AM : MB = 1 : 1$
3. So $A'M' : M'B' = 1 : 1$.

ANSWER

At the midpoint of $A'B'$

EXAMPLE 2 **A square is projected. Can its image be a rhombus that is not a square?**

1. A square is a parallelogram.
2. Any parallelogram can be the image of a square.
3. A rhombus is a parallelogram.

ANSWER

Yes

EXAMPLE 3 **Can the projection of a triangle be a segment?**

1. If the plane of the triangle is parallel to the direction of projection…

2. …every point drops onto one line.

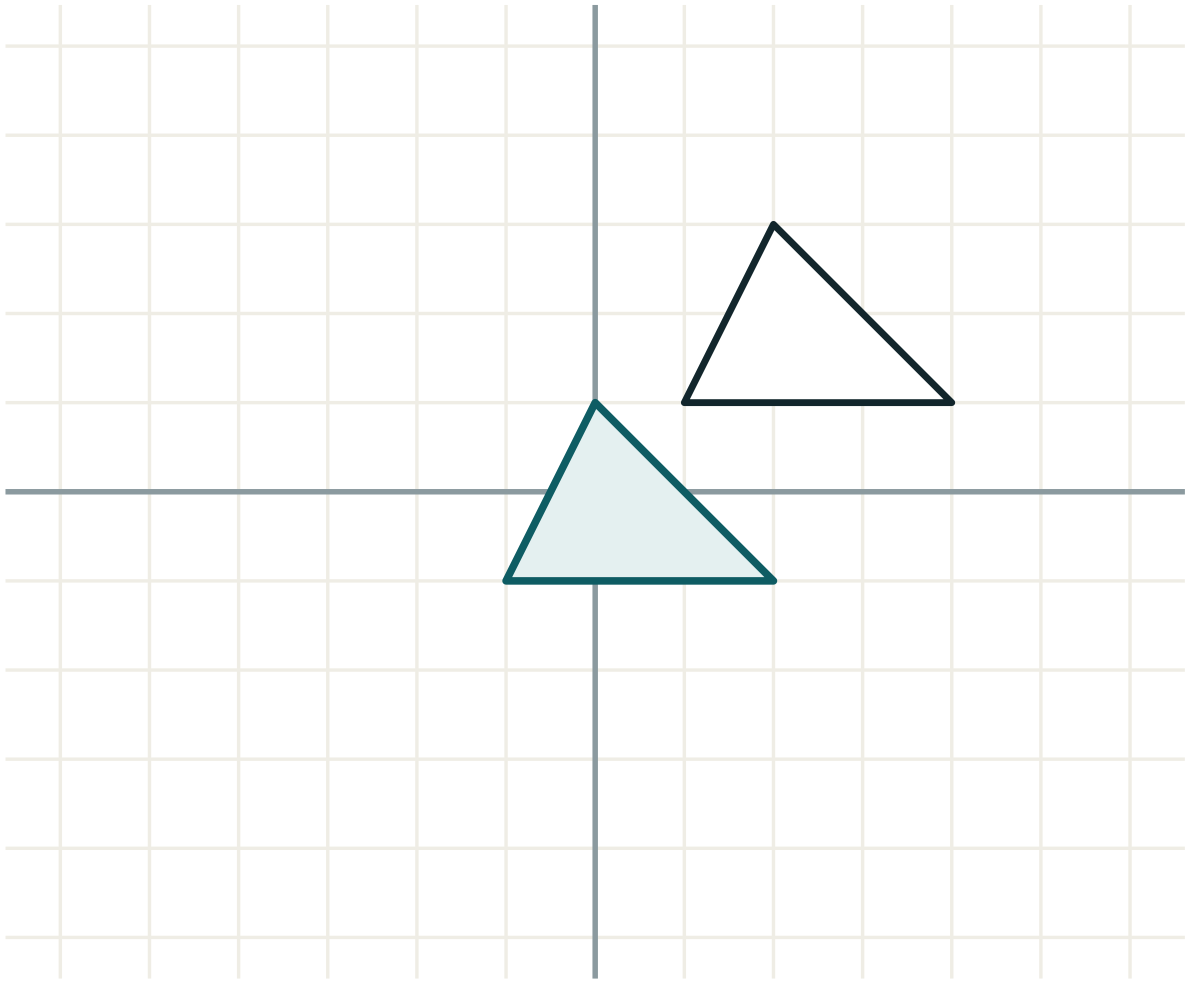
ANSWER

Yes — when the triangle's plane contains the direction of projection

03 Interactive model

Hold a wire square in sunlight and turn it — the shadow runs through every parallelogram.

What a projection preserves



transformation **translation** | lengths **unchanged** | angles **unchanged** | orientation **kept**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Parallel projection	Parallel proyeksiya	Параллельное проектирование
Direction of projection	Proyeksiyalash yo‘nalishi	Направление проектирования
Projecting line	Proyeksiyalovchi chiziq	Проектирующая прямая
Image (projection)	Proyeksiya	Проекция
Plane of projection	Proyeksiya tekisligi	Плоскость проекции
Preserved property	Saqlanuvchi xossa	Сохраняющееся свойство
Ratio of division	Bo‘linish nisbati	Отношение деления
Visible and hidden edges	Ko‘rinadigan va ko‘rinmaydigan qirralar	Видимые и невидимые рёбра
Correct picture	To‘g‘ri tasvir	Правильное изображение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Parallel projection preserves:

- lengths
- angles
- the midpoint
- areas

The image of a circle is:

- a circle
- an ellipse
- a parabola
- a line

The image of a square can be:

- only a square
- any parallelogram
- any quadrilateral
- only a rhombus

Angles under parallel projection are:

- preserved
- doubled
- not preserved
- halved

Hidden edges in a correct picture are:

- omitted
- dashed
- thickened
- coloured

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Image of a straight line

ANSWER a line (or a point)

2. Image of parallel lines

ANSWER parallel lines

3. Image of a midpoint

ANSWER the midpoint

4. Image of a circle

ANSWER an ellipse

5. Image of a square

ANSWER a parallelogram

6. Are lengths preserved?

ANSWER no

7. Are angles preserved?

ANSWER no

Medium · the standard question

7 PROBLEMS

1. Can a square project to a rhombus?

   ANSWER yes

2. Can a rectangle project to a trapezium?

   ANSWER no — parallel stays parallel

3. Can a triangle project to a segment?

   ANSWER yes

4. Is the ratio $AM : MB$ preserved?

   ANSWER yes

5. Is the ratio of two perpendicular edges preserved?

   ANSWER no

6. In cabinet projection the depth axis is drawn at

   ANSWER $45°$ and half scale

7. Why is the base of a cube drawn as a parallelogram?

   ANSWER A square projects to a parallelogram

Hard · stretch and reason

7 PROBLEMS

1. Prove that parallel projection preserves the midpoint

   ANSWER The projecting lines cut two parallels proportionally

2. Prove that parallel lines project to parallel lines

   ANSWER Their projecting planes are parallel

3. Can two perpendicular lines project to parallel lines?

   ANSWER Yes — if their common plane contains the direction

4. A cube of side 1 in cabinet projection: length of the receding edge on paper

   ANSWER 0.5

5. Why can any triangle be the image of any triangle?

   ANSWER An affine map takes any triangle to any other

6. Which is preserved: the ratio of areas, or the ratio of lengths on non-parallel lines?

   ANSWER the ratio of areas

7. Explain why a drawing must never be measured

   ANSWER Lengths and angles are not preserved

07 Homework

Homework — 5 tasks

Task 5 must be drawn, with hidden edges dashed.

1. List four properties preserved by parallel projection and three that are not.
2. Explain why a square can project to any parallelogram.
3. Can the projection of a rectangle be a trapezium? Justify your answer.
4. A point divides a segment in the ratio $2 : 3$. What is the ratio on the image?
5. Draw a correct picture of a cube and of a hexagonal prism, with hidden edges dashed.